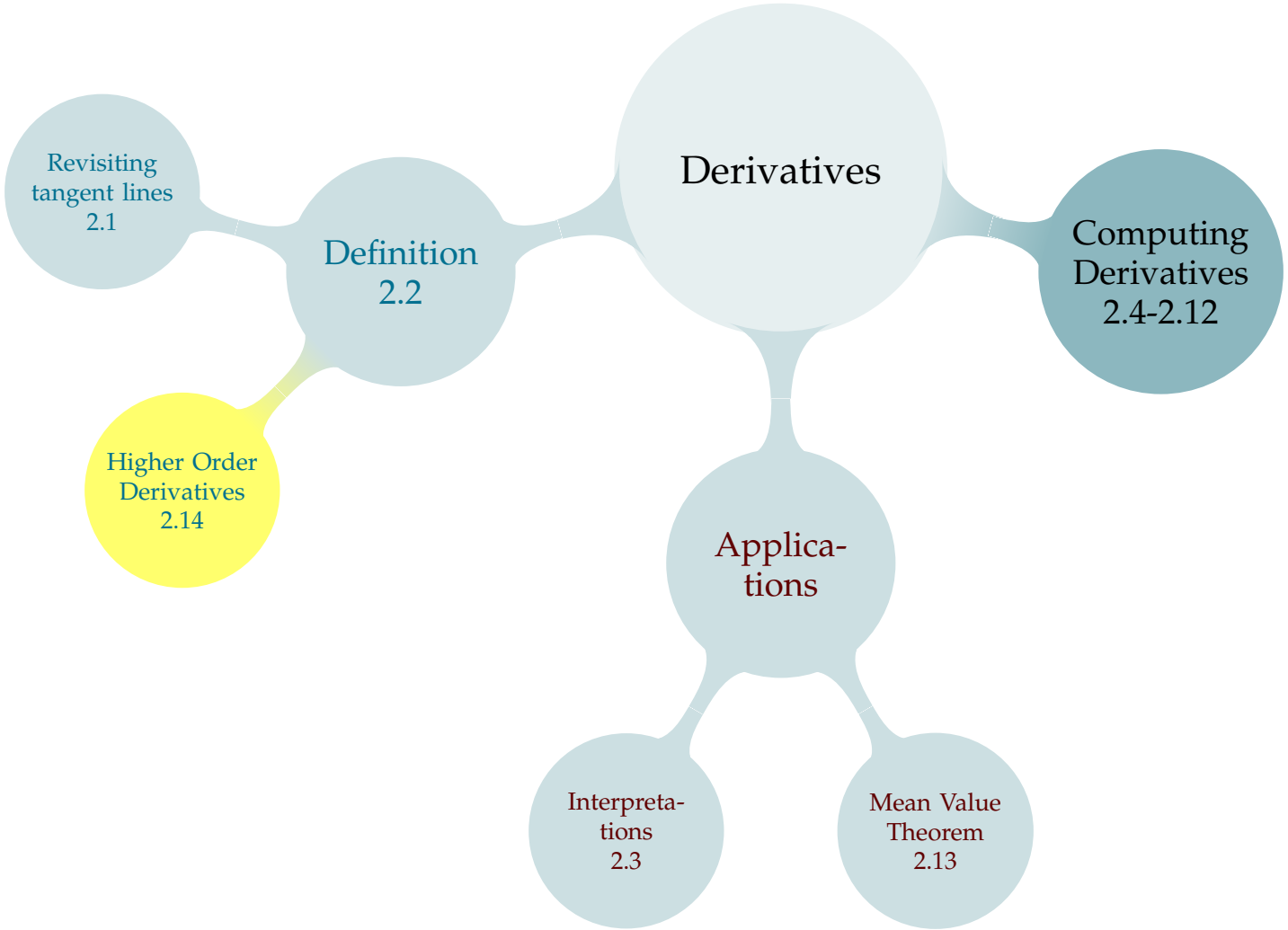
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ans

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Similarly, the derivative of a second derivative is a third derivative, etc.

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- $f''(x)$ and $f^{(2)}(x)$ and $\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}(x)$ all mean $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}f(x)\right)$
- $f'''(x)$ and $f^{(3)}(x)$ and $\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}(x)$ all mean $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}f(x)\right)\right)$
- $f^{(4)}(x)$ and $\frac{\mathrm{d}^4 f}{\mathrm{d}x^4}(x)$ both mean $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}f(x)\right)\right)\right)$
- and so on.

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- Velocity: rate of change of position.

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The position of an object at time t is given by $s(t) = t(5 - t)$. *Time is measured in seconds, and position is measured in metres.*

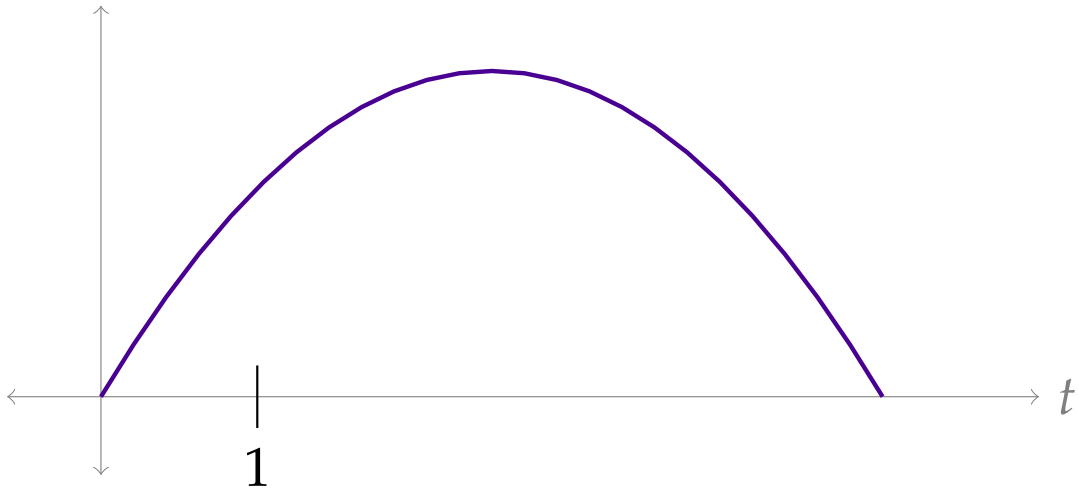
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$$s(t) = t(5 - t) = 5t - t^2$$
$$s'(t) = 5 - 2t$$
$$s'(1) = 5 - 2(1) = 3 = \text{vel}$$

Units of velocity: $\frac{\Delta s}{\Delta t} = \frac{m}{s}$ Units of acceleration: $\frac{\Delta s'}{\Delta t} = \frac{m/s}{s} = \frac{m}{s^2}$

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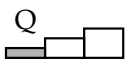
$$s'(t) = 5 - 2t \qquad s''(t) = -2$$

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True or False: If $f'(1) = 18$, then $f''(1) = 0$, since the $\frac{\mathrm{d}}{\mathrm{d}x}\{18\} = 0$.



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Which of the following is always true of a QUADRATIC polynomial $f(x)$?

A. $f(0) = 0$
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D. $f'''(0) = 0$
E. $f^{(4)}(0) = 0$

Q

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A. $f(0) = 0$ $f(x) = ax^2 + bx + c$
B. $f'(0) = 0$ $f'(x) = 2ax + b$
C. $f''(0) = 0$ $f''(x) = 2a$
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Which of the following is always true of a CUBIC polynomial $f(x)$?

A. $f(0) = 0$ $f(x) = ax^3 + bx^2 + cx + d$
B. $f'(0) = 0$ $f'(x) = 3ax^2 + 2bx + c$
C. $f''(0) = 0$ $f''(x) = 6ax + 2b$
D. $f'''(0) = 0$ $f'''(x) = 6a$
E. $f^{(4)}(0) = 0$ $f^{(4)}(x) = 0$ ✓

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Implicit Differentiation

Suppose $y(x)$ is a function such that

$$y(x) = y^3 x + x^2 - 1$$

Find $y''(x)$ at the point $(-2, 1)$.

Implicit Differentiation

Suppose $y(x)$ is a function such that

$$y(x) = y^3 x + x^2 - 1$$

Find $y''(x)$ at the point $(-2, 1)$. We start by differentiating both sides of the function. Remember that y is a function, not a variable.

$$y(x) = y(x)^3 x + x^2 + 1$$

$$\frac{dy}{dx}(x) \overset{\text{prod}}{=} y(x)^3 + 3xy(x)^2 \frac{dy}{dx}(x) + 2x \qquad (*)$$

Let's differentiate both sides again. Remember we have a rule for the product of three functions.

$$\frac{d^2y}{dx^2} = 3y^2 \frac{dy}{dx} + 3\left(y^2 \frac{dy}{dx} + x \cdot 2y \frac{dy}{dx} \cdot \frac{dy}{dx} + xy^2 \frac{d^2y}{dx^2}\right) + 2 \qquad (**)$$

When $x = -2$ and $y = 1$, using $(*)$, we find

$$\left.\frac{dy}{dx}\right|_{(-2,1)} = 1^3 + 3(-2)(1^2)\left.\frac{dy}{dx}\right|_{(-2,1)} + 2(-2) = -3 - 6\left.\frac{dy}{dx}\right|_{(-2,1)}$$

$$\left.\frac{dy}{dx}\right|_{(-2,1)} = -\frac{3}{7}$$

We set $x = -2$, $y = 1$, and $\frac{dy}{dx} = -\frac{3}{7}$ in equation $(**)$. Now by $\frac{d^2y}{dx^2}$, we actually mean $\left.\frac{d^2y}{dx^2}\right|_{(-2,1)}$, but to avoid clutter we don't write it that way until the end.

$$\begin{aligned}
\frac{d^2y}{dx^2} &= 3(1)\left(-\frac{3}{7}\right) + 3\left((1)\left(-\frac{3}{7}\right) + (-2)\cdot 2(1)\left(-\frac{3}{7}\right)\cdot\left(-\frac{3}{7}\right) + (-2)(1)\frac{d^2y}{dx^2}\right) + 2 \\
&= -\frac{9}{7} + 3\left(-\frac{3}{7} - \frac{36}{49} - 2\frac{d^2y}{dx^2}\right) + 2 \\
&= \left(2 - \frac{18}{7} - \frac{108}{49}\right) - 6\frac{d^2y}{dx^2} \\
7\frac{d^2y}{dx^2} &= -\frac{136}{7^2} \\
\left.\frac{d^2y}{dx^2}\right|_{(-2,1)} &= -\frac{136}{7^3}
\end{aligned}$$

Included Work


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